IT 308 – ONLINE PERFORMANCE TASK

Innovating the Future: Emerging Technology Application Design

Phase 1: Technology Selection

Chosen Technology:
Blockchain

Supporting Technology:
Mobile Money

Proposed App Title:
ReliefChain PH: Transparent Disaster Aid Distribution System

Phase 2: Application Concept

1. App Name

ReliefChain PH

2. Problem Statement

The Philippines frequently experiences typhoons, floods, and earthquakes. Southern Leyte, in particular, is prone to landslides and typhoon damage. During disasters, the government and private organizations send relief funds and donations. However, many issues arise such as lack of transparency, delayed distribution, corruption, and misallocation of funds.

Sometimes, beneficiaries do not receive the correct amount of aid, or records are not properly tracked. This creates distrust among citizens. There is a need for a secure, transparent, and tamper-proof system that ensures disaster funds are properly distributed to the right beneficiaries.

3. Technology Explanation

What is Blockchain?

Blockchain is a decentralized digital ledger technology that records transactions securely across multiple computers. Once data is recorded in a block and added to the chain, it cannot be altered or deleted. This ensures transparency, security, and trust.

How does it work?

1. A transaction is created (release of disaster funds).
2. The transaction is verified by a network of computers (nodes).
3. The verified transaction is stored in a block.
4. The block is added permanently to the chain.
5. Everyone in the network can view the record, but no one can change it.

Because blockchain is decentralized, no single person can manipulate the data. This makes it ideal for tracking financial transactions.

4. Application Features

- **Transparent Fund Tracking Dashboard**
  Citizens can view real-time records of donated funds and where they are allocated.
- **Verified Beneficiary Registration**
  Disaster victims are registered using valid ID verification to avoid duplication or fraud.
- **Smart Contract-Based Fund Release**
  Funds are automatically released once conditions are met (e.g., LGU approval, damage assessment completion).
- **Mobile Money Direct Transfer**
  Aid is sent directly to beneficiaries through mobile wallets to prevent middlemen corruption.
- **Audit and Reporting System**
  Government agencies and NGOs can generate detailed financial reports anytime.

5. Target Users

- Disaster victims in Southern Leyte
- Local Government Units (LGUs)

- NGOs and private donors
- National disaster response agencies
- Citizens monitoring public funds

6. Community Impact

ReliefChain PH promotes transparency and accountability in disaster fund distribution. It reduces corruption because transactions cannot be altered once recorded. It ensures that victims receive financial aid directly and quickly.

For Southern Leyte, where natural disasters are common, this system can improve trust between citizens and the government. It also modernizes disaster response systems in the Philippines. In the long term, it strengthens good governance and financial transparency nationwide.

7. Simple System Architecture Diagram

Donor / Government

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Mobile App / Web Portal

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Blockchain Network

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Smart Contract System

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Mobile Money Wallet

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Beneficiary (User)

Simplified Version:

User → App → Blockchain Network → Smart Contract → Mobile Wallet → Output

Phase 3: Enhancement Requirement

Blockchain + Mobile Money Integration

ReliefChain PH integrates:

- Blockchain (for transparency and security)
- Mobile Money (for fast and direct financial distribution)

Why is this combination effective?

Blockchain ensures that all transactions are recorded permanently and cannot be manipulated. However, blockchain alone does not deliver the funds physically to beneficiaries. This is where mobile money becomes important.

Mobile money allows direct digital transfer of funds to beneficiaries' wallets. When combined, blockchain ensures transparency, while mobile money ensures speed and accessibility.

Together, they create a secure, fast, and corruption-resistant disaster aid system.

Phase 4: Reflection

I chose Blockchain technology because transparency and corruption issues are serious concerns in the Philippines, especially during disaster relief operations. As a BSIT student, I believe technology should solve real societal problems, not just create convenience. Blockchain is powerful because it builds trust in digital systems.

One challenge in developing this system is the complexity of blockchain implementation. It requires advanced technical knowledge and strong cybersecurity measures. Another challenge is digital literacy, especially among disaster victims who may not be familiar with mobile money systems. Internet connectivity in rural areas can also be a limitation.

Despite these challenges, I believe this application is realistic. Many countries are already using blockchain for financial transparency and digital transactions. The Philippines is also rapidly adopting digital payment systems. With proper government support and infrastructure improvement, this system can be implemented nationwide.

To build this in the future, I need to improve my skills in blockchain development, smart contract programming, and cybersecurity. I also need stronger knowledge in database systems and mobile application development. Learning about financial technology systems would also be helpful.

As a future IT professional, I want to design systems that create social impact. This project reflects my goal of using technology to promote transparency, accountability, and trust in our country.